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Multi-slot PDCCH monitoring and search space set group switching

Abstract

A user equipment (UE) configured to receive configuration information for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM), receive configuration information for a second SS set of a second slot group for MSM, perform MSM based on the first SS set and switch from the first SS set to the second SS set using one of a first type of search space set group (SSSG) switching or a second type of SSSG switching.

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Background/Summary

BACKGROUND (1) This application claims priority to U.S. Provisional Application Ser. No. 63/267,849 filed on Feb. 11, 2022 and entitled “Multi-Slot PDCCH Monitoring and Search Space Set Group Switching,” the entirety of which is incorporated herein by reference.

BACKGROUND
(1) In a Fifth Generation (5G) New Radio (NR) network, for communication above 52.6 Giga hertz (GHz), the subcarrier spacing (SCS) may be increased to provide robustness to phase noise. For example, the SCS may be set to 120 kilo hertz (kHz), 480 kHz or 960 kHz. However, increasing the SCS may result in a reduction in the duration of the symbol which may place an unreasonable strain on user equipment (UE) processing resources during physical downlink control channel (PDCCH) monitoring. It has been identified that multi-slot PDCCH monitoring may be used to improve the efficiency of PDDCH monitoring for communication above 52.6 GHz.
SUMMARY
(2) Some exemplary embodiments are related to a processor of a user equipment (UE) configured

to perform operations. The operations include receiving configuration information for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM), receiving configuration information for a second SS set of a second slot group for MSM, performing MSM based on the first SS set and switching from the first SS set to the second SS set using one of a first type of search space set group (SSSG) switching or a second type of SSSG switching.
(3) Other exemplary embodiments are related to a processor of a base station configured to perform operations. The operations include transmitting configuration information to a user equipment (UE) for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM) and transmitting configuration information to the UE for a second SS set of a second slot group for MSM, wherein the UE is configured to perform search space set group (SSSG) switching between the first SS and the second SS.
(4) Still further exemplary embodiments are related to a method performed by a user equipment (UE). The method includes receiving configuration information for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM), receiving configuration information for a second SS set of a second slot group for MSM, performing MSM based on the first SS set and switching from the first SS set to the second SS set using one of a first type of search space set group (SSSG) switching or a second type of SSSG switching.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows an exemplary set of slot groups within a subframe according to various exemplary embodiments.
(2) FIG. **2** shows an example arrangement of Group 1 and Group 2 slot groups according to various exemplary embodiments.
(3) FIG. **3** shows an exemplary network arrangement according to various exemplary embodiments.
(4) FIG. **4** shows an exemplary user equipment (UE) according to various exemplary embodiments.
(5) FIG. **5** shows an exemplary base station according to various exemplary embodiments.
(6) FIG. **6** shows an example of search space (SS) set group (SSSG) switching according to various exemplary embodiments.
(7) FIG. **7** shows an example of SSSG switching according to various exemplary embodiments.
DETAILED DESCRIPTION
(8) The exemplary embodiments may be further understood with reference to the following description and the related appended drawings, wherein like elements are provided with the same reference numerals. The exemplary embodiments relate to enabling multi-slot physical downlink control channel (PDCCH) monitoring (MSM).
(9) The exemplary embodiments are described with regard to a user equipment (UE). However, reference to a UE is merely provided for illustrative purposes. The exemplary embodiments may be utilized with any electronic component that may establish a connection to a network and is configured with the hardware, software, and/or firmware to exchange information and data with the network. Therefore, the UE as described herein is used to represent any electronic component.
(10) In a Fifth Generation (5G) New Radio (NR) network, for communication above 52.6 Giga hertz (GHz), the subcarrier spacing (SCS) may be increased to provide robustness to phase noise. For example, the SCS may be set to 120 kilo hertz (kHz), 480 kHz or 960 kHz. However, increasing the SCS may result in a reduction in the duration of the symbol. From the perspective of the UE, the reduction in symbol duration may increase the number of operations that are to be

performed by the UE for PDCCH monitoring which may place an unreasonable strain on UE processing resources.
(11) It has been identified that it may be beneficial to utilize MSM for communication above 52.6 GHz. MSM may allow the UE to avoid the processing strain associated with other PDDCH monitoring approaches. However, while the exemplary techniques described herein may provide benefits to 5G NR communication above 52.6 GHz, the exemplary embodiments are not limited to this frequency range. Moreover, the exemplary embodiments may also be applied to future releases of the cellular standards, e.g., 6G.
(12) MSM may generally refer to a PDCCH monitoring approach that is based on slot groups that each comprise multiple consecutive slots. The UE may perform PDCCH monitoring within a search space (SS) during one or more slots of each slot group. To provide a general example, if a slot group comprises 4 consecutive slots, the UE may perform PDCCH monitoring during a SS within one or more slots of the 4 consecutive slots of the slot group. The UE may be configured with multiple slot groups that are each associated with the same or different frequency resources and/or overlap (fully or partially) in the time domain.
(13) The exemplary embodiments relate to SS set group (SSSG) switching. Those skilled in the art will understand that the UE may be configured with multiple different SS sets for PDCCH monitoring. To differentiate between different SS sets, each SS set may correspond to a different group index (e.g., 0, 1, etc.). The network may provide the UE with the group index for the different SS sets in a SS group ID list information element (IE). During operation, the UE may switch between different SSSGs for PDCCH monitoring.
(14) As will be described in more detail below, the exemplary embodiments introduce techniques related to the interaction between release 17 (Rel-17) SSSG switching mechanisms and release 16 (Rel-16) SSSG switching mechanisms. Each of the exemplary techniques may be used independently from one another, in conjunction with currently implemented MSM mechanisms, in conjunction with future implementations of MSM mechanisms or independently from other MSM mechanisms. Multi-Slot PDCCH Monitoring General Framework
(15) FIG. **1** shows an exemplary set of slot groups **140-144** within a subframe **130** according to various exemplary embodiments. This exemplary slot group arrangement is not intended to limit the exemplary embodiments in any way and is merely provided as a general overview of the relationship between a slot group and a subframe. A subframe may comprise 1 slot or multiple slots (e.g., 2, 4, 5, 12, 16, etc.) and the exemplary embodiments are not limited to any particular number of slots or slot groups per subframe.
(16) The UE **110** may be configured with a PDCCH that includes multiple subframes **130**. In this example, the PDCCH may be configured with a SCS of 480 KHz and 32 slots per subframe. FIG. **1** shows a portion of a subframe **130** with 12 slots indexed 0-11. This portion of subframe **130** is arranged into 3 separate slot groups **140-144** and each slot group **140-144** comprises 4 slots. There are 32 slots per subframe in this example and thus, the remaining portion of subframe **130** that is not pictured in FIG. **1** may include 20 slots indexed 12-31. The slots indexed 12-31 may be arranged into 5 separate groups each comprising a slot group size of 4 slots. Therefore, while only 3 slot groups **140-144** are shown in FIG. **1**, subframe **130** may include a total 8 slot groups with a slot group size of 4 slots across its 32 slots.
(17) In this example, the UE **110** may be configured to perform PDCCH monitoring in 1 slot from each of the slot groups **140-144**. Thus, in a first slot group **140** comprising slots indexed 0-3, the UE **110** may have a PDCCH SS during slot 1. During slots indexed 0, 2 and 3, the UE **110** has the opportunity to conserve power since the UE **110** is not configured to perform PDCCH monitoring during the other slots 0, 2, 3. In a second slot group **142** comprising slots indexed 4-7, the UE **110** may have a PDCCH SS during slot 5. During slots indexed 4, 6 and 7, the UE **110** has the opportunity to conserve power since the UE **110** is not configured to perform PDCCH monitoring during the other slots 4, 6, 7. In a third slot group **144** comprising slots indexed 8-11, the UE **110**

may have a PDCCH SS during slot 9. During slots indexed 8, 10 and 11, the UE **110** has the opportunity to conserve power since the UE **110** is not configured to perform PDCCH monitoring during the other slots 8, 10, 11. The UE **110** may behave in the same manner on the other 5 slot groups referenced above in the remaining portion of subframe **130** that is not pictured in FIG. **1**.
(18) Slot groups may be consecutive to one another. Thus, in this example, slot group **140** comprises slots indexed 0-3, slot group **142** comprises slots indexed 4-7 and slot group **144** comprises slots indexed 8-11. The start of a first slot group in a subframe (e.g., slot group **140**) may be aligned with a slot boundary (e.g., slot index 0 (not pictured)). The start of each slot group may be aligned with a slot boundary. In this example, there is no gap between the slot groups **140-144**. FIG. **1** is not intended to limit the exemplary embodiments in any way and is merely provided as a general overview of the relationship between a slot group and a subframe. The exemplary embodiments may apply to any appropriate SCS, subframe duration, number of slots per subframe, number of slot groups, slot group size, etc.
(19) A control resources set (CORESET) may be defined and based on the CORESET a SS may be defined. The UE **110** may perform PDCCH monitoring within the SS. The following examples provide a general overview of SSs within the slot group framework.
(20) Throughout this description, reference is made to "Group 1" to identify a first set of consecutive slot groups and "Group 2" to identify a second set of consecutive slot groups. Those skilled in the art will understand that MSM Group 1 and Group 2 may be defined in third generation partnership program (3GPP) Specifications. The exemplary embodiments may utilize Group 1 and Group 2 in accordance with the manner in which these terms are defined in 3GPP documents and may be modified in accordance with the exemplary embodiments described herein.
(21) The UE **110** may perform PDCCH monitoring within a SS configured within one or more slots of each slot group. For Group 1, the SSs may include a type 1 common search space (CSS) with dedicated radio resource control (RRC) configuration, a type 3 CSS, a UE specific SS and/or any other appropriate type of SS. For Group 2, the SSs may include a type 1 CSS without dedicated RRC configuration, a type 0 CSS, a type OA CSS, a type 2 CSS and/or any other appropriate type of SS. Thus, "Group 1" and "Group 2" may encompass different types of SSs. Specific details regarding the arrangement of a Group 1 SS set and a Group 2 SS set are provided below.
(22) The slot group size for Group 1 may be the same as or different than the slot group size for Group 2. In addition, Group 1 and Group 2 may each be associated with the same or different frequency resources and overlap (fully or partially) in the time domain. However, any reference to a particular Group 1 and/or Group 2 arrangement is not intended to limit the exemplary embodiments in any way and is merely provided as an example. The exemplary embodiments may apply to any appropriate SCS, subframe duration, number of slots per subframe, number of slot groups, slot group size, etc.
(23) Group 1 may consist of (Xs) consecutive slots and a SS may be configured within (Ys) consecutive slots within the Xs slots of the slot group. The location of the Ys slots within the slot group may be maintained across different slot groups. To provide an example within the context of FIG. **1**, Xs may be equal to 4 (e.g., slot index 0-3) and Ys may be equal to 1. The position of the SS (e.g., Ys) in slot group **140** is the same position of the SS in slot group **142** and **144**.
(24) When Ys is equal to 1, the SS may be located anywhere within the Ys slot. However, the SS location may be subject to a gap-span limitation (W, Z) (e.g., release 15 (rel-15) gap-span, feature group (FG) 3-5b). For 480 KHz, the gap-span limitation (W, Z) may be (4, 3) or (7, 3) with a maximum of two monitoring spans per the Ys slot. For 960 KHs, the gap-span limitation (W, Z) may be (7, 3). An example arrangement of a Group 1 slot group is provided below with regard to FIG. **2**.
(25) When Ys is greater than 1, the SSs may be located in the first 3 symbols of each of the Ys slots. To provide an example within the context of FIG. **1**, if each slot comprised symbols indexed 1-14, the SS in slot 1 of slot group **140** may be located within symbols indexed 1-3. Similarly, the

SS in slot 1 of slot group **142** may be located within symbols indexed 1-3 and the SS in slot 1 of slot group **144** may be located within symbols indexed 1-3.

(26) Group 2 may consist of (Xs) consecutive slots and a Group 2 SS may be configured anywhere within the Xs consecutive slots. However, there may be some exceptions such as, but not limited to, type 0 CSS with multiplex pattern 1 and type OA/2 CSS with a SS ID equal to 0, where the location of the SS within the slot group is based on a particular parameter (e.g., time offset, symbol index, etc.) and/or derived based on a particular equation. An example arrangement of a Group 2 slot group is provided below with regard to FIG. **2**.

(27) FIG. **2** shows an example arrangement of Group 1 and Group 2 according to various exemplary embodiments. The example arrangement depicted in FIG. **2** is not intended to limit the exemplary embodiments in any way and is merely provided to demonstrate a general example of how the exemplary MSM framework described above may be utilized.

(28) In this example, Group 1 is configured with a slot group that comprises (Xs=4) slots. Thus, slot group **210** includes slots **211**, **212**, **213** and **214**. Each slot 211-214 may comprise 14 symbols indexed 0-13. Similarly, Group 2 is also configured with a slot group that comprises (Xs=4) slots. Thus, Slot group **250** includes slots **251**, **252**, **253** and **254**. Each slot 251-254 may comprise 14 symbols indexed 0-13. In this example, the arrangement of slots in Group 1 and Group 2 are the same. However, as mentioned above, the exemplary embodiments are not limited to this arrangement and may utilize any appropriate SCS, subframe duration, number of slots per subframe, number of slot groups, slot group size, etc.

(29) In addition, Group 1 is configured with a Group 1 SS within (Ys=2) slots. In this example, the Ys slots include slot 212 and slot 213. Since Ys is greater than 1, the SS **220** in slot 212 and the SS **222** in slot 213 may be located within the first 3 symbols of each slot (e.g., symbols indexed 0-2). Group 2 is configured with SSs in three slots **251-253**. Since a Group 2 SS may be located anywhere within a slot group, the SSs **262**, **264** and **266** are shown as being located within a different span of symbols for each of the slot groups **251-253**. However, the exemplary embodiments are in no way limited to this Group 2 SS set configuration. This is just one possible example of a Group 2 SS set configuration and the exemplary embodiments may apply to any appropriate Group 2 configuration.

(30) Network Arrangement and Components

(31) FIG. **3** shows an exemplary network arrangement **300** according to various exemplary embodiments. The exemplary network arrangement **300** includes the UE **110**. Those skilled in the art will understand that the UE **110** may be any type of electronic component that is configured to communicate via a network, e.g., mobile phones, tablet computers, desktop computers, smartphones, phablets, embedded devices, wearables, Internet of Things (IoT) devices, etc. It should also be understood that an actual network arrangement may include any number of UEs being used by any number of users. Thus, the example of a single UE **110** is merely provided for illustrative purposes.

(32) The UE **110** may be configured to communicate with one or more networks. In the example of the network arrangement **300**, the network with which the UE **110** may wirelessly communicate is a 5G NR radio access network (RAN) **320**. However, the UE **110** may also communicate with other types of networks (e.g., 5G cloud RAN, a next generation RAN (NG-RAN), a long-term evolution (LTE) RAN, a legacy cellular network, a wireless local area network (WLAN), etc.) and the UE **110** may also communicate with networks over a wired connection. With regard to the exemplary embodiments, the UE **110** may establish a connection with the 5G NR RAN **320**. Therefore, the UE **110** may have a 5G NR chipset to communicate with the 5G NR RAN **320**.

(33) The 5G NR RAN **320** may be a portion of a cellular network that may be deployed by a network carrier (e.g., Verizon, AT&T, T-Mobile, etc.). The 5G NR RAN **320** may include, for example, nodes, cells or base stations (e.g., Node Bs, eNodeBs, HeNBs, eNBS, gNBs, gNodeBs, macrocells, microcells, small cells, femtocells, etc.) that are configured to send and receive traffic

from UEs that are equipped with the appropriate cellular chip set.

(34) Those skilled in the art will understand that any association procedure may be performed for the UE **110** to connect to the 5G NR PAN **320**. For example, as discussed above, the 5G NR PAN **320** may be associated with a particular cellular provider where the UE **110** and/or the user thereof has a contract and credential information (e.g., stored on a SIM card). Upon detecting the presence of the 5G NR PAN **320**, the UE **110** may transmit the corresponding credential information to associate with the 5G NR PAN **320**. More specifically, the UE **110** may associate with a specific base station, e.g., the next generation Node B (gNB) **320**A.

(35) The network arrangement **300** also includes a cellular core network **330**, the Internet **340**, an IP Multimedia Subsystem (IMS) **350**, and a network services backbone **360**. The cellular core network **330** may refer an interconnected set of components that manages the operation and traffic of the cellular network. It may include the evolved packet core (EPC) and/or the 5G core (5GC). The cellular core network **330** also manages the traffic that flows between the cellular network and the Internet **340**. The IMS **350** may be generally described as an architecture for delivering multimedia services to the UE **110** using the IP protocol. The IMS **350** may communicate with the cellular core network **330** and the Internet **340** to provide the multimedia services to the UE **110**. The network services backbone **360** is in communication either directly or indirectly with the Internet **340** and the cellular core network **330**. The network services backbone **360** may be generally described as a set of components (e.g., servers, network storage arrangements, etc.) that implement a suite of services that may be used to extend the functionalities of the UE **110** in communication with the various networks.

(36) FIG. **4** shows an exemplary UE **110** according to various exemplary embodiments. The UE **110** will be described with regard to the network arrangement **300** of FIG. **3**. The UE **110** may include a processor **405**, a memory arrangement **410**, a display device **415**, an input/output (I/O) device **420**, a transceiver **425** and other components **430**. The other components **430** may include, for example, an audio input device, an audio output device, a power supply, a data acquisition device, ports to electrically connect the UE **110** to other electronic devices, etc.

(37) The processor **405** may be configured to execute a plurality of engines of the UE **110**. For example, the engines may include a MSM engine **435**. The MSM engine **435** may perform various operations related to MSM including, but not limited to, receiving MSM parameters, identifying a location of one or more Group 1 SSs, identifying a location of one or more Group 2 SSs and monitoring MSM. In addition, the MSM engine **435** may implement the various exemplary techniques introduced herein related to Rel-16 and Rel-17 SSSG switching interaction.

(38) The above referenced engine **435** being an application (e.g., a program) executed by the processor **405** is merely provided for illustrative purposes. The functionality associated with the engine **435** may also be represented as a separate incorporated component of the UE **110** or may be a modular component coupled to the UE **110**, e.g., an integrated circuit with or without firmware. For example, the integrated circuit may include input circuitry to receive signals and processing circuitry to process the signals and other information. The engines may also be embodied as one application or separate applications. In addition, in some UEs, the functionality described for the processor **405** is split among two or more processors such as a baseband processor and an applications processor. The exemplary embodiments may be implemented in any of these or other configurations of a UE.

(39) The memory arrangement **410** may be a hardware component configured to store data related to operations performed by the UE **110**. The display device **415** may be a hardware component configured to show data to a user while the I/O device **420** may be a hardware component that enables the user to enter inputs. The display device **415** and the I/O device **420** may be separate components or integrated together such as a touchscreen. The transceiver **425** may be a hardware component configured to establish a connection with the 5G NR-RAN **320**, an LTE-RAN (not pictured), a legacy RAN (not pictured), a WLAN (not pictured), etc. Accordingly, the transceiver

425 may operate on a variety of different frequencies or channels (e.g., set of consecutive frequencies).
(40) FIG. **5** shows an exemplary base station **500** according to various exemplary embodiments. The base station **500** may represent the gNB **320**A or any other access node through which the UE **110** may establish a connection and manage network operations.
(41) The base station **500** may include a processor **505**, a memory arrangement **510**, an input/output (I/O) device **515**, a transceiver **520**, and other components **525**. The other components **525** may include, for example, an audio input device, an audio output device, a battery, a data acquisition device, ports to electrically connect the base station **500** to other electronic devices, etc.
(42) The processor **505** may be configured to execute a plurality of engines for the base station **500**. For example, the engines may include a MSM engine **530**. The MSM engine **530** may perform various operations related to the UE **110** performing MSM including, but not limited to, signaling MSM parameters to the UE **110**, receiving capability information, configuring CSSs, configuring UE specific SSs, managing a BD/CCE budget and scheduling PDCCH resources for the UE **110**.
(43) The above noted engine **530** being an application (e.g., a program) executed by the processor **505** is only exemplary. The functionality associated with the engine **530** may also be represented as a separate incorporated component of the base station **500** or may be a modular component coupled to the base station **500**, e.g., an integrated circuit with or without firmware. For example, the integrated circuit may include input circuitry to receive signals and processing circuitry to process the signals and other information. In addition, in some base stations, the functionality described for the processor **505** is split among a plurality of processors (e.g., a baseband processor, an applications processor, etc.). The exemplary embodiments may be implemented in any of these or other configurations of a base station.
(44) The memory **510** may be a hardware component configured to store data related to operations performed by the base station **500**. The I/O device **515** may be a hardware component or ports that enable a user to interact with the base station **500**. The transceiver **520** may be a hardware component configured to exchange data with the UE **110** and any other UE in the network arrangement **300**. The transceiver **520** may operate on a variety of different frequencies or channels (e.g., set of consecutive frequencies). Therefore, the transceiver **520** may include one or more components (e.g., radios) to enable the data exchange with the various networks and UEs.
(45) Rel-16 and Rel-17 SSSG Switching Interaction
(46) As mentioned above, the exemplary embodiments introduce techniques related to the interactions between Rel-16 SSSG switching mechanisms and Rel-17 SSSG mechanisms. In Rel-17, a SSSG switching timer and PDCCH skipping feature were introduced as a UE power saving enhancement for operation in frequency range (FR)-2 with 480 and/or 960 kHz SCS. However, it has been identified that these Rel-17 features may conflict with Rel-16 DCI format 2_0 based SSSG switching. The exemplary embodiments introduce techniques for enabling the operation of these Rel-16 and Rel-17 features.
(47) The Rel-16 based SSSG switching mechanism may be configured with 2 SS sets where one of the SS sets is configured as a default SS set. The Rel-17 based SSSG switching mechanism may be configured with 3 SS sets where one of the SS sets is configured as a default SS set. Therefore, a scenario may occur where the UE **110** is configured with 5 SS sets with two different default SS sets, e.g., a Rel-16 default SS set and a Rel-17 default SS set.
(48) The Rel-16 SSSG switching may provide power saving benefits in the unlicensed band. For instance, the UE **110** may be provided with at least two groups of SS sets for PDCCH. The UE **110** may be configured to switch between the groups. Those skilled in the art will understand that using the unlicensed band may include performing listen before talk (LBT) to determine whether the channel is occupied prior to performing a transmission on the channel. In some examples, after LBT is successful, the UE **110** may be configured with a dense SS set to enable the UE **110** to find a start of a slot. An example of this mechanism is shown in FIG. **6** during a transmission burst

occupying one transmission opportunity (TXOP). In this example, after successful LBT and prior to a first full slot, a SS set (e.g., group index 1) may be used by the UE **110** to find the start of the first full slot. Subsequently, the UE **110** may switch to a sparser SS set (e.g., group index 2).

(49) In contrast, Rel-17 SSSG switching may provide power saving benefits to all bands (e.g., licensed, unlicensed, etc.). In some scenarios, the SSSG switching mechanism may be controlled by a SSSG switching timer operated by the UE **110**. The value of the timer may be configured by the network and indicate when the UE **110** is to switch from monitoring a first SS set to monitoring a second different SS set. An example of SSSG switching is shown in FIG. **7**. In this example, the UE **110** monitors a first SSSG **705** during time windows **710** and **715**. The UE **110** monitors a second SSSG **750** during time window **755**. The UE **110** may switch between monitoring the SSSG **705** and the SSSG **750** based on the SSSG switching timer or any other appropriate mechanism.

(50) As indicated above, the exemplary embodiments introduce techniques for enabling the operation of different SSSG switching mechanisms at the UE **110** (e.g., Rel-16 SSSG switching and Rel-17 SSSG switching). In one approach, Rel-16 SSSG switching and Rel-17 SSSG switching may not be simultaneously configured. Thus, while the UE **110** may support both of the SSSG switching mechanisms, a base station may be barred from and/or intentionally avoid configuring the UE **110** with a simultaneous configuration of Rel-16 SSSG switching and Rel-17 SSSG switching. For example, the gNB **320**A may identify that the UE **110** supports both of these mechanisms. The gNB **320**A may then configure the UE **110** with one of Rel-16 SSSG switching or Rel-17 SSSG switching at a first time. Subsequently, the gNB **320**A may configure the UE **110** with the other SSSG switching mechanism. However, the UE **110** intentionally avoids providing the UE **110** with a simultaneous configuration of Rel-16 SSSG switching and Rel-17 SSSG switching.

(51) In another approach, Rel-16 SSSG switching and Rel-17 SSSG switching may be simultaneously configured when NR-dual connectivity (DC) and/or carrier aggregation (CA) is configured at the UE **110**. However, the UE **110** does not expect to be simultaneously configured within a same band. For example, the network via one or more serving cells (e.g., primary cell (PCell), primary secondary cell (PSCell), secondary cell (SCell), etc.) may simultaneously configure the UE **110** with two different SSSG switching mechanisms (e.g., Rel-16 SSSG switching and Rel-17 SSSG switching). However, one of the SSSG switching mechanisms is configured for a first set of one or more frequency bands (or component carriers (CCs)) and the other SSSG switching mechanism is configured for a second different set of frequency bands (or CCs). Since the UE **110** does not expect to be simultaneously configured within a same band, the first set of frequency bands (e.g., unlicensed, etc.) may not include any frequency bands from the second set of frequency bands (e.g., licensed, etc.) or vice versa.

(52) In a further approach, a simultaneous configuration of both Rel-16 SSSG switching and Rel-17 SSSG switching may be utilized. Thus, in some examples, there may be 5 SSs configured at the same time. In this example, a single default SS may be defined. In some embodiments, a default SS may be based on Rel-16 SSSG configuration where a first SS is utilized after LBT and prior to a first full slot. In other embodiments, the default SS may be based on the Rel-17 default configuration where a first SS is used after a first slot but before the end of the channel occupancy time (COT). In another embodiment, a default SS may be set by the network (e.g., gNB **320**A). In contrast to the above example, the network may choose one of the configurations as the default SS and indicate to the UE **110** which SS is the default SSSG. Those skilled in the art will understand that the default SS is a SS that the UE **110** may be configured to monitor if there is no explicitly indicated SS. For example, in timer based switching, the UE **110** may be triggered to switch to a non-default group for a specific duration. When the duration expires, the UE **110** switches back to monitoring the default SS.

(53) In another approach, the network may configure a maximum of 3 SSSG configurations comprising both Rel-16 and Rel-17 SSs mapping to the three. For example, consider a scenario in

which the UE **110** is configured with timer based SSSG switching and a default SS. The UE **110** knows which SS to switch to for both Rel-16 and Rel-17 SSSG switching. Thus, when both Rel-16 and Rel-17 SSs are configured, the network (e.g., gNB **320**A) may dynamically switch between Rel-16 and Rel-17 SSSG switching. The non-default Rel-16 SSSG switching SS may be mapped to one of the other two configurations while up to two Rel-17 SSSG switching SSs may be mapped to the non-default configurations.

(54) In another approach, a mix of the Rel-16 SSSG switching mechanism and the Rel-17 SSSG switching mechanism may be implemented. With the approach, a monitoring group flag may be introduced in DCI format 2_0. This bit flag may be used to indicate a switch to a non-default Rel-17 SSSG SSs. For example, the UE **110** may receive configuration information for Rel-16 SSSGs and Rel-17 SSSGs. One of the mechanisms may be set as the default mechanism. For instance, the Rel-16 SSSG mechanism may be configured as a default mechanism. Thus, the UE **110** may initially utilize the default SSSG mechanism. During operation, the UE **110** may receive DCI format 2_0 with a bit flag set to indicate to the UE **110** that the UE **110** is to switch from the default SSSG mechanism to the other SSSG mechanism.

(55) In some embodiments, the bit flag is present in the DCI (e.g., explicit signaling). When the bit flag is set to a first value (e.g., 1), the UE **110** may switch (or keep) monitoring Rel-17 SSSG SSs at the next applicable slot boundary relative to the detected DCI format 2_0. In addition, the UE **110** may initiate a configurable timer. As will be explained in more detail below, the timer may be used to trigger the SSSG switching.

(56) When the bit flag is set to a second different value (e.g., 0), the UE **110** may switch (or keep) monitoring the default SS at the next appliable slot boundary relative to the detected DCI format 2_0.

(57) In some embodiments, the DCI format 2_0 may not contain a bit flag (e.g., implicit switching) or the UE **110** may not monitor for DCI format 2_0. In this type of scenario, if any PDCCH in the default SS is successfully detected, the UE **110** may switch from Group 1 to the Rel-17 SSs at the next applicable slot boundary relative to the detected PDCCH. The next applicable boundary may refer to the earliest start of a slot that is at least P1 or P2 symbols later than the last symbol of the corresponding PDCCH. The values P1 and P2 may not be less than the processing time required for the UE **110** to perform the SS switching.

(58) The UE **110** may initiate the timer in response to PDCCH (e.g., DCI or any other appropriate type of PDCCH). The UE **110** may switch from Rel-17 SSSG SSs to Group 1 at the earliest slot boundary that is at least P2 symbols after the end of the slot that coincides with the expiration of the timer or an indicated COT duration is exceeded.

(59) In another example, for a frequency band with LBT (e.g., unlicensed, etc.), Rel-17 default SS may be configured for the UE **110** outside of an LBT region. When LBT starts, a Rel-16 default SS may be used. In addition, a CSS may be used at a start of a slot boundary.

(60) Those skilled in the art will understand that the above-described exemplary embodiments may be implemented in any suitable software or hardware configuration or combination thereof. An exemplary hardware platform for implementing the exemplary embodiments may include, for example, an Intel x86 based platform with compatible operating system, a Windows OS, a Mac platform and MAC OS, a mobile device having an operating system such as iOS, Android, etc. The exemplary embodiments of the above described methods may be embodied as a program containing lines of code stored on a non-transitory computer readable storage medium that, when compiled, may be executed on a processor or microprocessor.

(61) Although this application described various embodiments each having different features in various combinations, those skilled in the art will understand that any of the features of one embodiment may be combined with the features of the other embodiments in any manner not specifically disclaimed or which is not functionally or logically inconsistent with the operation of the device or the stated functions of the disclosed embodiments.

(62) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.
(63) It will be apparent to those skilled in the art that various modifications may be made in the present disclosure, without departing from the spirit or the scope of the disclosure. Thus, it is intended that the present disclosure cover modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalent.

Claims

1. A processor of a user equipment (UE) configured to perform operations comprising: receiving configuration information for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM); receiving configuration information for a second SS set of a second slot group for MSM; performing MSM based on the first SS set; and switching from the first SS set to the second SS set using one of a first type of search space set group (SSSG) switching or a second type of SSSG switching.
2. The processor of claim 1, wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching and wherein Rel-16 SSSG switching and Rel-17 SSSG switching are not simultaneously configured at the UE.
3. The processor of claim 1, wherein the UE is configured with one of new radio (NR)-dual connectivity (DC) or carrier aggregation (CA), wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching, and wherein Rel-16 SSSG switching is configured within an unlicensed band and Rel-17 SSSG switching is configured within a licensed band simultaneously.
4. The processor of claim 1, wherein the UE is configured with one of new radio (NR)-dual connectivity (DC) or carrier aggregation (CA), wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching, and wherein Rel-16 SSSG switching and Rel-17 SSSG switching are not simultaneously configured within a same band.
5. The processor of claim 1, wherein the UE is configured with five different SS sets, wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching, and wherein Rel-16 SSSG switching and Rel-17 SSSG switching are simultaneously configured.
6. The processor of claim 5, wherein a single default configuration for SSSG switching is based on a Rel-16 SSSG configuration.
7. The processor of claim 5, wherein a single default configuration for SSSG switching is based on a Rel-17 SSSG configuration.
8. The processor of claim 5, the operations further comprising: receiving a signal from a base station indicating a default configuration for SSSG switching.
9. The processor of claim 1, the operations further comprising: receiving downlink control indication (DCI) indicating that the UE is to perform MSM using a release 16 (Rel-16) SS during listen before talk (LBT) and release 17 (Rel-17) SS at a next slot boundary relative to the LBT.
10. A processor of a base station configured to perform operations comprising: transmitting configuration information to a user equipment (UE) for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM); and transmitting configuration information to the UE for a second SS set of a second slot group for

MSM, wherein the UE is configured to perform search space set group (SSSG) switching between the first SS and the second SS.

11. The processor of claim 10, wherein the base station configures three SS sets for the UE with both release 16 (Rel-16) and release 17 (Rel-17) SS sets mapping to the three SS sets.

12. The processor of claim 11, wherein a single default configuration is set for both the Rel-16 SS sets and the Rel-17 SS sets.

13. The processor of claim 12, wherein a Rel-17 SS is mapped to a non-default configuration.

14. The processor of claim 10, the operations further comprising: transmitting downlink control indication (DCI) to the UE indicating that the UE is to perform MSM using a release 17 (Rel-17) SS at a next slot boundary relative to the DCI.

15. The processor of claim 10, the operations further comprising: transmitting downlink control indication (DCI) to the UE indicating that the UE is to perform MSM using a default SS at a next slot boundary relative to the DCI.

16. A method performed by a user equipment (UE), comprising: receiving configuration information for a first search space (SS) set of a first slot group for multi-slot physical downlink control channel (PDCCH) monitoring (MSM); receiving configuration information for a second SS set of a second slot group for MSM; performing MSM based on the first SS set; and switching from the first SS set to the second SS set using one of a first type of search space set group (SSSG) switching or a second type of SSSG switching.

17. The method of claim 16, wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching and wherein Rel-16 SSSG switching and Rel-17 SSSG switching are not simultaneously configured at the UE.

18. The method of claim 16, wherein the UE is configured with one of new radio (NR)-dual connectivity (DC) or carrier aggregation (CA), wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching, and wherein Rel-16 SSSG switching is configured within an unlicensed band and Rel-17 SSSG switching is configured within a licensed band simultaneously.

19. The method of claim 16, wherein the UE is configured with one of new radio (NR)-dual connectivity (DC) or carrier aggregation (CA), wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching, and wherein Rel-16 SSSG switching and Rel-17 SSSG switching are not simultaneously configured within a same band.

20. The method of claim 16, wherein the UE is configured with five different SS sets, wherein the first type of SSG switching is release 16 (Rel-16) SSSG switching and the second type of SSG switching is release 17 (Rel-17) SSSG switching, and wherein Rel-16 SSSG switching and Rel-17 SSSG switching are simultaneously configured.